

Biometric Authentication Solutions Comparison

A comprehensive comparison of SABLE, AWS Rekognition, and native on-device biometrics (Android/iOS) for identity verification and authentication.

Executive Summary

Solution	Primary Use Case	Privacy Model	Network Required
SABLE	Prove identity to third parties without exposing biometrics	Zero-knowledge proofs	No (fully offline)
AWS Rekognition	Face detection, analysis, and recognition at scale	Cloud processing	Yes
Android/iOS Biometrics	Authenticate user to their own device	On-device only	No

Detailed Feature Comparison

Core Capabilities

Feature	SABLE	AWS Rekognition	Android/iOS Biometrics
Authenticate to own device	✔	✘	✔
Prove identity to third parties	✔	✔	✘
Works offline	✔	✘	✔
Cross-device verification	✔	✔	✘
P2P identity verification	✔	✘	✘
Government PKI integration	✔	✘	✘

Biometric Modalities

Modality	SABLE	AWS Rekognition	Android/iOS Biometrics
Face recognition	✔	✔	✔ (Face ID/Face Unlock)
Palm vein/print	✔	✘	✘
Fingerprint	Extensible	✘	✔ (Touch ID/Fingerprint)
Iris	Extensible	✘	✔ (Samsung Iris)
Emotion detection	✘	✔	✘
Facial landmarks	✘	✔	✘
Age/attribute analysis	✘	✔	✘

Privacy & Security

Aspect	SABLE	AWS Rekognition	Android/iOS Biometrics
Biometric data location	Device only	AWS cloud	Device secure enclave
Data sent to servers	Never (only ZK proofs)	Yes (images/video)	Never
Central database	None	Optional Face Collections	None
Breach exposure risk	Minimal	Cloud storage risk	Minimal
Government access	Cryptographically prevented	Possible via AWS	Device-locked
Privacy model	Zero-knowledge proofs	Trust AWS	Local-only

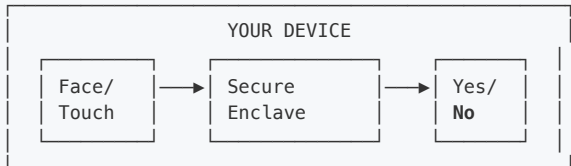
Technical Architecture

Aspect	SABLE	AWS Rekognition	Android/iOS Biometrics
Processing location	On-device	Cloud	Secure Enclave/TEE
Cryptography	BLS12-381, Groth16 zk-SNARKs	Proprietary ML	Hardware-backed keys
Proof size	192 bytes	N/A	N/A
Verification time	<2 seconds	Network-dependent	<1 second
API type	Rust library + FFI	REST API	Platform SDK

The Fundamental Difference

On-Device Biometrics (Android/iOS): "Unlock Your Phone"

Native biometrics answer one question: "Is this the device owner?"



⚠ Cannot prove anything **to external parties**

Limitations:

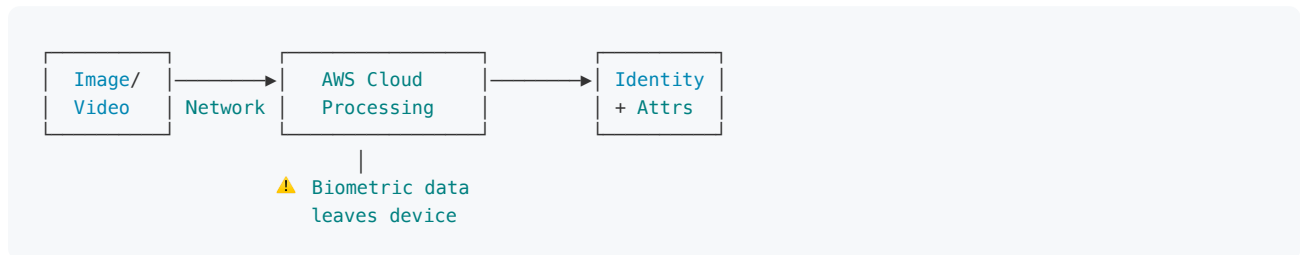
- Only works on YOUR device with YOUR enrolled biometrics
- Cannot prove your identity to a website, another person, or government
- If you lose your device, you lose your identity credential
- Third parties must trust your device's attestation (not your identity)

Use cases:

- Unlocking your phone
- Authorizing Apple Pay/Google Pay (device attests, not you)
- Accessing banking apps on your device
- Password manager unlock

AWS Rekognition: "Who Is This Face?"

Rekognition answers: "Whose face is in this image/video?"



Limitations:

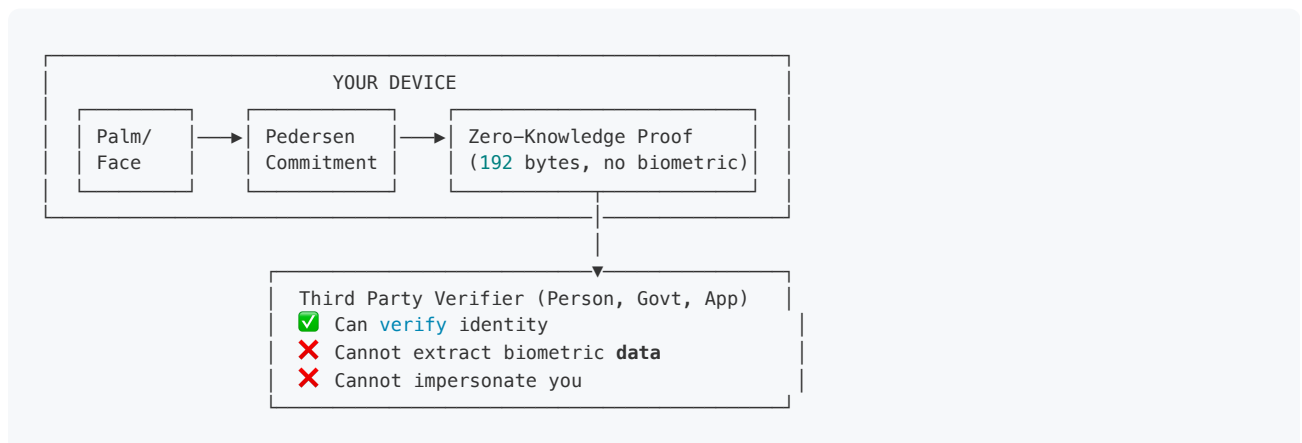
- Requires internet connectivity
- Biometric data processed on AWS servers
- Pay-per-API-call pricing
- Privacy concerns for sensitive applications
- No offline capability

Use cases:

- Security camera person identification
- Photo organization and tagging
- Content moderation
- Attendance systems (with privacy tradeoffs)
- Celebrity recognition

SABLE: "Prove You Are You, Without Revealing Yourself"

SABLE answers: "Can you prove your identity without exposing your biometrics?"



Unique capabilities:

- Prove identity to anyone, anywhere, offline
- Biometric data mathematically cannot be extracted from proofs
- Government can issue credentials without storing biometrics
- P2P verification without central authority
- Works across devices (credential portability)

Use cases:

- Age verification without revealing birthdate
- Building access without physical cards
- Government services without biometric surveillance
- P2P trust establishment
- Cross-border identity verification
- Privacy-preserving KYC

Scenario Comparison

Scenario 1: Unlocking Your Phone

Solution	Capability	Notes
Android/iOS Biometrics	✔ Best choice	Native, fast, hardware-backed
SABLE	✔ Possible	Overkill for this use case
AWS Rekognition	✗ Not designed for this	Cloud latency, requires network

Scenario 2: Proving Age (18+) at a Bar

Solution	Capability	Notes
Android/iOS Biometrics	✗ Cannot do this	Only proves device ownership
SABLE	✔ Best choice	Proves age without revealing DOB or biometrics
AWS Rekognition	⚠ Possible	Requires network, exposes face to cloud

Scenario 3: Building Access Control

Solution	Capability	Notes
Android/iOS Biometrics	⚠ Limited	Device must be pre-registered; can't verify identity
SABLE	✔ Best choice	Offline P2P verification, no central database
AWS Rekognition	⚠ Possible	Requires network, privacy concerns

Scenario 4: Government Benefits Verification

Solution	Capability	Notes
Android/iOS Biometrics	✗ Cannot do this	Cannot prove identity to government
SABLE	✓ Best choice	Government attests without storing biometrics
AWS Rekognition	⚠ Possible	Creates government biometric database

Scenario 5: Analyzing Emotions in Customer Videos

Solution	Capability	Notes
Android/iOS Biometrics	✗ Cannot do this	Not designed for analysis
SABLE	✗ Not designed for this	Privacy-focused, not analytics
AWS Rekognition	✓ Best choice	Rich emotion and attribute detection

Cost Comparison

Reference Deployment: 5 Million Users

All cost projections are based on a **5 million user deployment** with **1 verification per user per day**.

Metric	Value
Total users	5,000,000
Verifications per user/day	1
Daily verifications	5,000,000
Monthly verifications	150,000,000
Annual verifications	1,825,000,000

Pricing Models Overview

Solution	Pricing Model	Base Cost
SABLE	Open source (Apache 2.0)	\$0 license fee
AWS Rekognition	Pay-per-API-call	\$0.001/image (tier 1)
Android/iOS Biometrics	Platform-included	\$0

AWS Rekognition Detailed Pricing

Image Analysis (Face Operations)

Volume (monthly)	Price per Image	Monthly Cost at Tier
First 1M images	\$0.001	\$1,000
Next 4M images	\$0.0008	\$3,200
Next 30M images	\$0.0006	\$18,000
Over 35M images	\$0.0004	Variable

Storage Costs

Item	Price
Face metadata storage	\$0.00001/face/month
5M faces stored	\$50/month

Free Tier (First 12 Months Only)

- 1,000 images/month — negligible for production

Monthly Cost Calculation (150M Verifications)

AWS Rekognition Monthly Breakdown

Tier	Images	Rate	Cost
Tier 1	1,000,000	\$0.001	\$1,000
Tier 2	4,000,000	\$0.0008	\$3,200
Tier 3	30,000,000	\$0.0006	\$18,000
Tier 4	115,000,000	\$0.0004	\$46,000
Total API	150,000,000		\$68,200
Face storage	5,000,000	\$0.00001	\$50
Monthly Total			\$68,250

SABLE Monthly Breakdown

Component	Cost
License	\$0
API calls	\$0
Cloud infrastructure	\$0

Component	Cost
Data transfer	\$0
Monthly Total	\$0

Android/iOS Biometrics

Component	Cost	Notes
License	\$0	
API calls	\$0	
Monthly Total	\$0	⚠️ Cannot verify identity to third parties

Annual Cost Comparison (5M Users)

Cost Component	SABLE	AWS Rekognition	Android/iOS
API/Transaction fees	\$0	\$818,400	\$0
Face storage	\$0	\$600	\$0
Data transfer (est.)	\$0	\$108,000	\$0
Annual Total	\$0	\$927,000	\$0

Note: Android/iOS biometrics cannot prove identity to third parties, making it unsuitable for this use case.

5-Year Total Cost of Ownership (5M Users)

Cost Component	SABLE	AWS Rekognition	Difference
API/Transaction fees	\$0	\$4,092,000	\$4,092,000
Face storage	\$0	\$3,000	\$3,000
Data transfer	\$0	\$540,000	\$540,000
Infrastructure/DevOps	\$0	\$300,000	\$300,000
Compliance & audit	\$50,000	\$500,000	\$450,000
Security monitoring	\$25,000	\$250,000	\$225,000
Breach liability reserve	\$0	\$1,000,000	\$1,000,000
Development & integration	\$150,000	\$200,000	\$50,000
5-Year Total	\$225,000	\$6,885,000	\$6,660,000

Cost Per User Per Year

Solution	Annual Cost	Cost per User/Year
SABLE	\$45,000	\$0.009
AWS Rekognition	\$1,377,000	\$0.275

SABLE is 30x cheaper per user per year.

Scaling Scenarios

Cost at Different User Counts (1 verification/user/day, annual)

Users	Verifications/Year	SABLE	AWS Rekognition	Savings
100,000	36.5M	\$0	\$18,980	\$18,980
500,000	182.5M	\$0	\$82,125	\$82,125
1,000,000	365M	\$0	\$158,775	\$158,775
5,000,000	1.825B	\$0	\$818,400	\$818,400
10,000,000	3.65B	\$0	\$1,606,800	\$1,606,800
50,000,000	18.25B	\$0	\$7,738,800	\$7,738,800
100,000,000	36.5B	\$0	\$15,331,800	\$15,331,800

Cost at Different Verification Frequencies (5M users, annual)

Frequency	Verifications/Year	SABLE	AWS Rekognition
1x/week	260M	\$0	\$116,400
1x/day	1.825B	\$0	\$818,400
2x/day	3.65B	\$0	\$1,606,800
5x/day	9.125B	\$0	\$3,898,800
10x/day	18.25B	\$0	\$7,738,800

Hidden Costs Deep Dive

Cost Category	SABLE	AWS Rekognition	Notes
License fees	\$0	\$0	Both free to start
Per-transaction	\$0	\$0.0004-\$0.001	Adds up at scale
Data transfer	\$0	\$0.09/GB out	~150M images × 100KB = 15TB/mo

Cost Category	SABLE	AWS Rekognition	Notes
Face storage	\$0	\$50/mo (5M faces)	Ongoing cost
Network bandwidth	\$0	User pays	Mobile data costs
Infrastructure	\$0	AWS account + ops	DevOps overhead
Compliance (GDPR/CCPA)	Minimal	Significant	Privacy by design vs. configuration
Security audits	\$10K/year	\$100K/year	More attack surface with cloud
Breach insurance	Low premium	High premium	No biometric database = low risk
Breach notification	N/A	\$50-\$150/record	5M records = \$250M-\$750M exposure

Risk-Adjusted Cost Analysis

Risk Factor	SABLE	AWS Rekognition
Probability of breach (5yr)	<1%	5-10%
Records at risk	0	5,000,000
Cost per breached record	N/A	\$150 avg
Expected breach cost	\$0	\$37.5M - \$75M
Risk-adjusted 5yr TCO	\$225,000	\$44M - \$82M

Device Resource Costs (SABLE)

While SABLE has no API fees, there are device-level costs:

Resource	Per Verification	Daily (1 verify)	Notes
Battery	0.03%	0.03%	Negligible
Memory	128MB peak	Temporary	Released after
CPU time	~2 seconds	~2 seconds	Background OK
Storage	~50MB	One-time	Library + creds

Estimated device cost per verification: < \$0.0001 (battery wear)

Annual device cost per user: ~\$0.04 (365 × \$0.0001)

Cost Summary (5M Users, 1 Verify/Day)

Metric	SABLE	AWS Rekognition
Monthly API cost	\$0	\$68,200

Metric	SABLE	AWS Rekognition
Annual API cost	\$0	\$818,400
5-Year TCO	\$225,000	\$6,885,000
Cost per verification	\$0	\$0.00045
Cost per user/year	\$0.009	\$0.275
5-Year savings	—	\$6,660,000

Break-Even Analysis

Comparison	Break-Even Point
SABLE vs Rekognition (API only)	1 verification (SABLE always wins)
SABLE vs Rekognition (with \$150K dev)	333M verifications (~67 days at 5M users)
Rekognition payback vs SABLE	Never (SABLE remains cheaper)

When Rekognition Costs Make Sense

Despite higher costs, AWS Rekognition may be justified when:

- ✔ You need emotion/attribute analysis (SABLE doesn't offer this)
- ✔ Volume is low (<1,000/month) and you use free tier
- ✔ You're already in AWS ecosystem with negotiated pricing
- ✔ You need celebrity recognition or content moderation
- ✔ Development time is more valuable than API costs

Cost Summary

Solution	Best For	5-Year Cost (1M/mo)
SABLE	High-volume, privacy-critical	~\$60,000
AWS Rekognition	Analytics, managed service	~\$783,000
Android/iOS	Device-only auth	~\$10,000

Security Considerations

Threat: Biometric Database Breach

Solution	Risk Level	Mitigation
SABLE	✔ None	No central database exists

Solution	Risk Level	Mitigation
AWS Rekognition	⚠️ Medium	AWS security + encryption
Android/iOS	✅ Minimal	Secure enclave isolation

Threat: Government Surveillance

Solution	Risk Level	Mitigation
SABLE	✅ Protected	Zero-knowledge proofs prevent access
AWS Rekognition	⚠️ Possible	Data stored on AWS servers
Android/iOS	✅ Protected	Local-only, no network transmission

Threat: Replay Attacks

Solution	Protection
SABLE	30-second time windows + 256-bit challenge nonces
AWS Rekognition	Application-dependent
Android/iOS	Hardware-backed freshness guarantees

Threat: Device Compromise

Solution	Risk Level	Notes
SABLE	⚠️ Medium	Rooted devices could extract templates
AWS Rekognition	✅ Low	Server-side processing
Android/iOS	⚠️ Medium	Jailbreak/root compromises enclave

Integration Complexity

Aspect	SABLE	AWS Rekognition	Android/iOS
SDK availability	Rust + FFI (Android/iOS)	REST API + SDKs	Native platform SDK
Setup time	Hours	Minutes	Minutes
Learning curve	Moderate (crypto concepts)	Low	Low
Documentation	Growing	Extensive	Extensive
Community support	Open source community	AWS support	Platform support

Decision Framework

Choose SABLE when:

- ☒ Privacy is non-negotiable
- ☒ Biometrics must never leave the device
- ☒ Offline operation is required
- ☒ Need to prove identity to third parties
- ☒ Government integration without surveillance
- ☒ P2P verification scenarios
- ☒ No ongoing cloud costs desired

Choose AWS Rekognition when:

- ☒ Need emotion/attribute/landmark analysis
- ☒ Processing large volumes of images/videos
- ☒ Building monitoring/surveillance systems
- ☒ Content moderation at scale
- ☒ Okay with cloud processing
- ☒ Need managed infrastructure

Choose Android/iOS Biometrics when:

- ☒ Only need to authenticate device owner
- ☒ Authorizing payments or app access
- ☒ Simple unlock/authentication flows
- ☒ Don't need to prove identity to external parties
- ☒ Want simplest possible integration

Summary Table

Requirement	Best Solution
Unlock my phone	Android/iOS Biometrics
Authorize a payment	Android/iOS Biometrics
Prove my age to a bartender	SABLE
Prove my identity to government	SABLE
Verify someone's identity offline	SABLE
Analyze emotions in videos	AWS Rekognition
Search for faces in video archive	AWS Rekognition
Build access without cards	SABLE
Content moderation	AWS Rekognition
Privacy-preserving KYC	SABLE

Conclusion

These three solutions serve fundamentally different purposes:

1. **Android/iOS Biometrics:** Best for **local device authentication**. Simple, fast, secure, but cannot prove your identity to anyone else.
2. **AWS Rekognition:** Best for **face analysis and recognition at scale**. Rich features for detection, emotion, and attributes, but requires sending biometric data to the cloud.
3. **SABLE:** Best for **privacy-preserving identity verification**. Unique ability to prove identity to third parties without ever exposing biometric data. Works fully offline with government-grade attestation.

The choice depends on your core requirement: **local auth** (native), **analysis at scale** (Rekognition), or **private identity proof** (SABLE).